Appendix of "Detection-Explanation-Improvement: A Closed-Loop Framework of Enhancing Anomaly Detection with Counterfactual Explanations"

1 Details of Bisection Search

The core iterative logic of the bisection method is as follows:

1. Initialize the step size search interval as $[\alpha_{\text{low}}, \alpha_{\text{high}}] = [0, 1]$.
2. Calculate the midpoint of the interval $\alpha_{\text{mid}} = (\alpha_{\text{low}} + \alpha_{\text{high}})/2$, and evaluate the anomaly score of the sample $\boldsymbol{x_a} + \alpha_{\text{mid}} \cdot \boldsymbol{p}_{\text{masked}}$.
3. If $f(\boldsymbol{x_a} + \alpha_{\text{mid}} \cdot \boldsymbol{p}_{\text{masked}}) \leq \tau$, it indicates that the current step size satisfies the constraint. Update the search interval to $[\alpha_{\text{low}}, \alpha_{\text{mid}}]$ and try to find a smaller step size.
4. If $f(\boldsymbol{x_a}+\alpha_{\text{mid}} \cdot \boldsymbol{p}_{\text{masked}}) > \tau$, it indicates that the step size is insufficient. Update the search interval to $[\alpha_{\text{mid}}, \alpha_{\text{high}}]$.
5. Repeat steps 2-4 until the length of the interval is less than the precision threshold ϵ, and take the left endpoint of the final interval as the optimal step size.

By solving the optimal step size with the bisection method, the feature adjustment magnitude can be minimized under the constraint of decision reversal, achieving the minimization goal of L_1 loss. Compared with linear search, the bisection method has a time complexity of $O(\log \frac{1}{\epsilon})$, where ϵ is the precision threshold of step size search. This enables efficient localization of the optimal step size across the decision boundary in logarithmic time.

2 Proof of Theorem 1

Given a fixed dataset X, we iteratively update the feature weight vector $w_{(t)}$. We aim to prove the following three key properties of the Separability Score sequence $\{\text{SS}_{(t)}\}_{t=0}^{\infty}$:

1. The sequence $\{\text{SS}_{(t)}\}_{t=0}^{\infty}$ is non-decreasing;
2. The sequence $\{\text{SS}_{(t)}\}_{t=0}^{\infty}$ is strictly bounded above;
3. By the Monotone Convergence Theorem, the sequence converges to a unique finite limit $\text{SS}^{\star}$.

Step 1: Proof of Non-decreasing Sequence

In each iteration, we only update the feature weights if the new candidate weight vector $w_{(t+1)}$ yields a higher Separability Score than the current one. If the candidate weights fail to improve the score, we retain the original weights and the Separability Score remains unchanged. Mathematically, the update rule is formulated as:

$$\text{SS}_{(t+1)} = \max\left\{\text{SS}_{(t)},\ \text{SS}(w_{(t+1)})\right\} \tag{1}$$

For any iteration round $t \geq 0$, the sequence thus satisfies the inequality:

$$\text{SS}_{(0)} \leq \text{SS}_{(1)} \leq \text{SS}_{(2)} \leq \cdots \leq \text{SS}_{(t)} \leq \text{SS}_{(t+1)} \leq \ldots$$

Conclusion: The sequence $\{\text{SS}_{(t)}\}_{t=0}^{\infty}$ is non-decreasing.

Step 2: Proof of Upper Bound for the Sequence

The Separability Score (SS) metric is formally defined as:

$$\text{SS} = \frac{\mu_{\text{a}} - \mu_{\text{n}}}{\sigma_{\text{n}} + \epsilon_0}, \tag{2}$$

where μ_{a} and μ_{n} denote the mean prediction scores of anomaly samples and normal samples at iteration t, respectively. σ_{n} is the standard deviation of prediction scores for normal samples, and $\epsilon_0 = 10^{-9}$ is a fixed positive smoothing term to avoid division by zero.

For anomaly detection models, the prediction score function $f(w_{(t)}, x)$ is universally bounded for a fixed dataset X. There exists a constant $M > 0$ (independent of iteration t and weight vector $w_{(t)}$ such that the prediction score of any sample $x \in X$ satisfies:

$$0 \leq f(w_{(t)}, x) \leq M.$$

Accordingly, the mean score of anomaly samples satisfies $\mu_{\text{a}(t)} \leq M$, and the mean score of normal samples satisfies $\mu_{\text{n}(t)} \geq 0$. The numerator of Eq. (2) is thus upper bounded by:

$$\mu_{\text{a}(t)} - \mu_{\text{n}(t)} \leq M - 0 = M.$$

By the fundamental mathematical property of standard deviation, the sample standard deviation is always non-negative for any iteration $t \geq 0$:

$$\sigma_{\text{n}(t)} = \sqrt{\mathbb{V}\text{ar}\left(f_{\text{n}(t)}\right)} \geq 0. \tag{3}$$

Combining with the fixed smoothing term $\epsilon_0 > 0$, the denominator of Eq. (2) has a positive lower bound:

$$\sigma_{\text{n}(t)} + \epsilon_0 \geq \epsilon_0 > 0.$$

By combining the upper bound of the numerator and the positive lower bound of the denominator, we derive the global upper bound of the Separability Score sequence:

$$\text{SS}_{(t)} = \frac{\mu_{\text{a}(t)} - \mu_{\text{n}(t)}}{\sigma_{\text{n}(t)} + \epsilon_0} \leq \frac{M}{\epsilon_0} = \mathcal{M}_{\text{max}},$$

where $\mathcal{M}_{\max} = M/\epsilon_0$ is a constant independent of iteration t.

Conclusion: The sequence $\{\mathrm{SS}_{(t)}\}_{t=0}^{\infty}$ is strictly bounded above.

Step 3: Final Convergence Proof

We have rigorously proven two core properties of the sequence $\{\mathrm{SS}_{(t)}\}_{t=0}^{\infty}$: it is non-decreasing and strictly bounded above. Based on the **Monotone Convergence Theorem** for real-valued sequences:

> If a real sequence is non-decreasing and bounded above, then it must converge to a unique finite limit.

Applying this theorem to the Separability Score sequence, we conclude that:

$$\lim_{t\to\infty} \mathrm{SS}_{(t)} = \mathrm{SS}^{\star},$$

where $\mathrm{SS}^{\star} \leq \mathcal{M}_{\max}$ is the unique finite limit of the sequence.

Final Conclusion: The iterative optimization process of the feature weight vector converges necessarily.